



MEMORANDUM

To: Professor Brendon Anderson, Mechanical Engineering, California Polytechnic State University, bga@calpoly.edu
From: Gabriel Choi, Carmen Alaichamy
Date: May 13, 2026
Subject: Lab 4: Tracking Control

Introduction

The objective of this lab was to implement a trapezoidal velocity profile for tracking control. The goal is to have the PI control match the given velocity profile in Figure 1. Finally, we will analyze the performance of the controller with a ramp input.

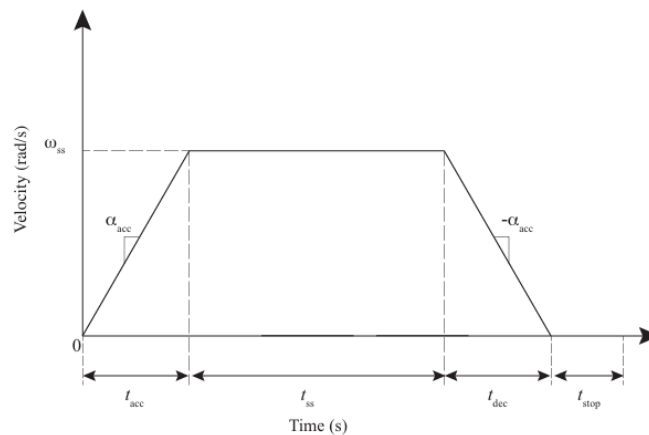


Figure 1. Given velocity profile.

Methodology

The experiment was conducted using an STM32 microcontroller connected to a DC motor and rotary encoder. We completed the file profile.py shown in Appendix A to create the velocity profile generator. We completed the generator function trapz_profile_gen() using equations based on the velocity profile graph. Then the method trapz_profile_length() computes the number of points needed. Running profile.py yields our desired velocity profile shown in Figure 2. The class file was then uploaded to the microcontroller using Thonny.

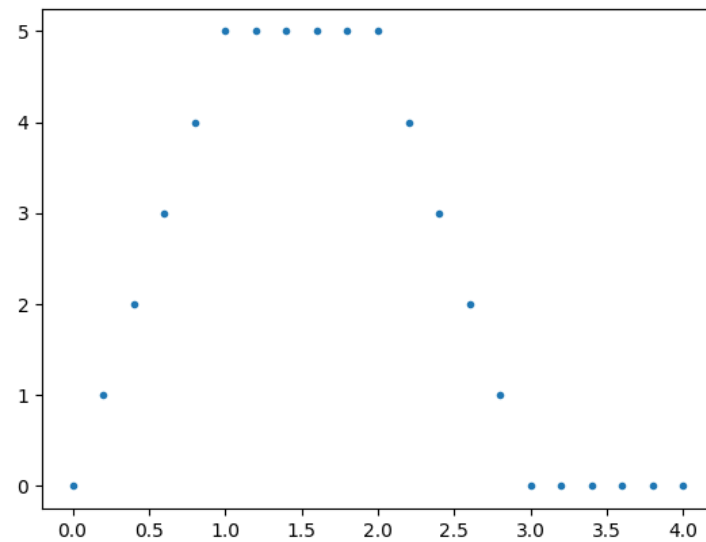


Figure 2. Profile.py velocity profile.

The main.py file is shown in Appendix B, it was completed to set the setpoint, run the controller, apply the correct voltage and collect data into queues. The main.py was tested and uploaded to the microcontroller.

To graph, the Jupyter Notebook file was used. This is shown in Appendix C, including the plots created.

Results

The following plots in Figure 3-7 shows the overlaid response plots of the ideal speed profile and the actual response. Figure 3 shows the gains from Lab 3b. Figure 4 and 5 modify these gains to

get the optimal response and an overshooting response, respectively. Figure 6 and 7 decrease acceleration to 0.25 seconds and show varying gains.

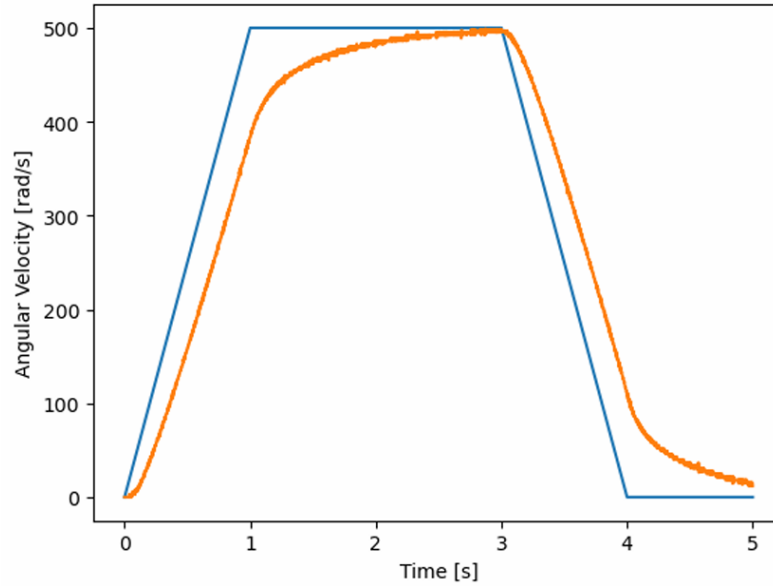


Figure 3. Overlaid response plot of *original* speed profile and actual response, *with controller gains from Lab 3B*. (Parameters: Steady state velocity = 500 rad/s, steady state time = 2 s, stop time of 1, acceleration time = 1 s, $K_p = 0.19$, $K_I = 0.4$)

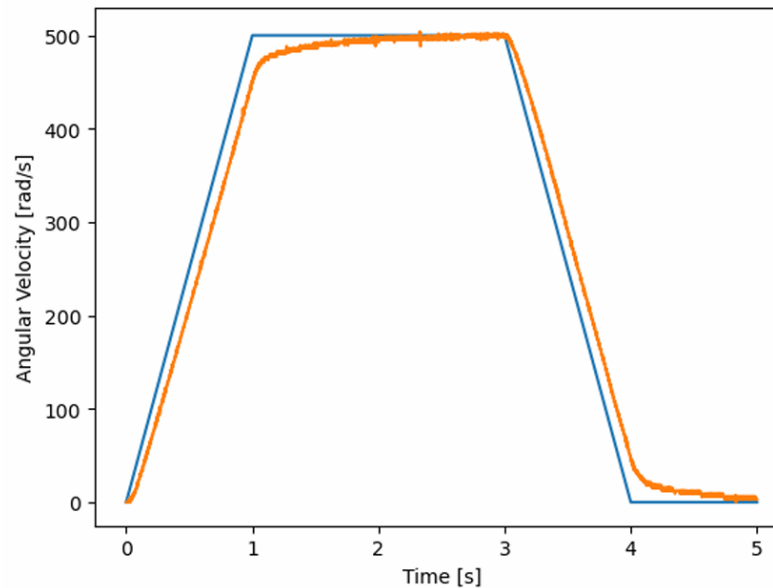


Figure 4. Overlaid response plot of *optimal* speed profile and actual response, *with modified controller gains*. (Parameters: Steady state velocity = 500 rad/s, steady state time = 2 s, stop time of 1, acceleration time = 1 s, $K_p = 0.5$, $K_I = 1.0$)

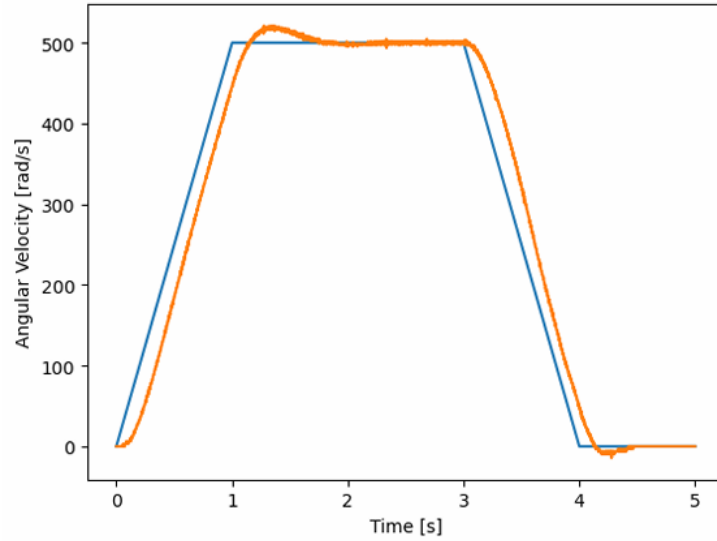


Figure 5. Overlaid response plot of *overshooting* speed profile and actual response, *with modified controller gains*. (Parameters: Steady state velocity = 500 rad/s, steady state time = 2 s, stop time of 1, acceleration time = 1 s, $K_p = 0.1$, $K_I = 1$)

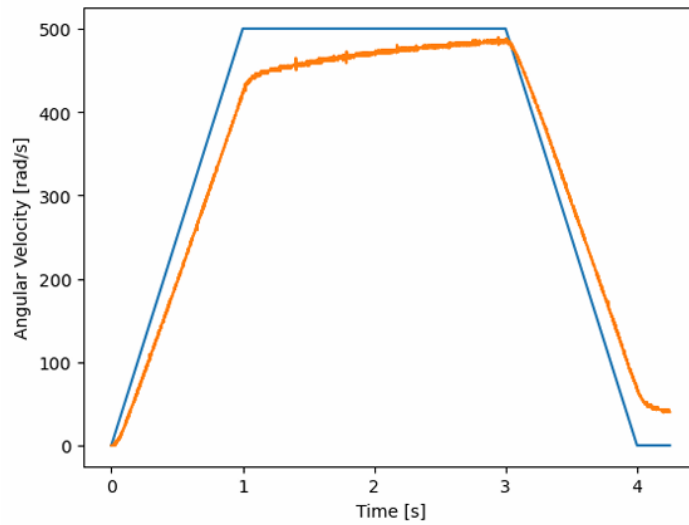


Figure 6. Overlaid response plot of speed profile and actual response, *with 0.25 acceleration and modified controller gains*. (Parameters: Steady state velocity = 500 rad/s, steady state time = 2 s, stop time of 1, acceleration time = 0.25 s, $K_p = 0.5$, $K_I = 0.4$)

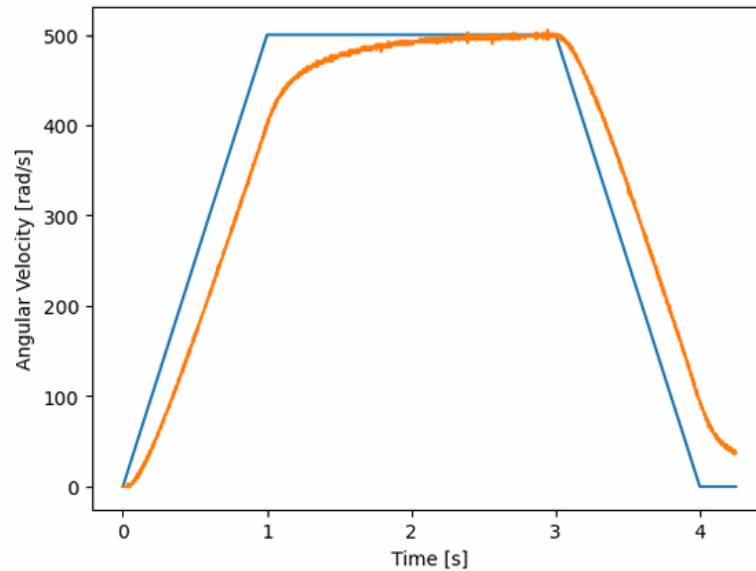


Figure 7. Overlaid response plot of speed profile and actual response, *with 0.25 acceleration and modified controller gains*. (Parameters: Steady state velocity = 500 rad/s, steady state time = 2 s, stop time of 1, acceleration time = 0.25 s, $K_p = 0.2$, $K_I = 0.5$)

Discussion

Explain how PI gains affect the ramp response. Is steady state error eliminated during the acceleration and deceleration of the motor?

The PI gains affect ramp response with each K_p and K_I contributing differently. K_p acts to quickly changing ramp response to decrease error. K_I integrates error over time to eliminate steady state error.

Between Figure 6 and 7, K_I is kept relatively constant while K_p was decreased from 0.5 to 0.2. The tracking in Figure 7 is clearly worse, with less K_p leading to slower response.

Steady state error is eliminated best in Figure 4, where K_I was increased to 1 from the previous 0.4 in Figure 3. Figure 3 reaches steady state with more error because there is less integral controller to recognize and fix the error.

Steady state error is eliminated most in Figure 4 because of the optimal balance between K_p and K_I controllers. While the steady state error becomes completely eliminated when the motor is at a constant speed, it is still not completely eliminated while the motor is accelerating or decelerating. The motor changing seems to make the controllers too slow to fully account for the error, or predict where the controller should operate.